

MAT 115: DISCRETE MATHEMATICS I LECTURE NOTES

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Chapter 1

INTRODUCTION

1.1 What is Discrete Mathematics?

Definition 1.1.1. The study of mathematical properties of sets and systems that have a countable number of elements.

Definition 1.1.2. Discrete mathematics is the branch of mathematics dealing with objects that can assume only distinct, separated values. The term "discrete mathematics" is therefore used in contrast with "continuous mathematics," which is the branch of mathematics dealing with objects that can vary smoothly (and which includes, for example, calculus). Whereas discrete objects can often be characterized by integers, continuous objects require real numbers.

Definition 1.1.3. Discrete mathematics is the study of mathematical structures that are fundamentally discrete in the sense of not supporting or requiring the notion of continuity....

1.2 Importance of studying Discrete Mathematics

- a) Discrete mathematics sharpens your mathematical thinking skills. Discrete math teaches mathematical reasoning and proof techniques.
- b) Discrete mathematics improves your problem solving abilities.

- c) Discrete mathematics is a foundation course to many other courses and disciplines.
- d) Discrete mathematics is fun.
- e) Discrete mathematics has become popular in recent decades because of its applications to computer science. Discrete mathematics is the mathematical language of computer science. Concepts and notations from discrete mathematics are useful in studying and describing objects and problems in all branches of computer science, such as computer algorithms, programming languages, cryptography, automated theorem proving, and software development.
- f) Discrete math plays the significant role in big data analytics. The Big Data era poses a critically difficult challenge and striking development opportunities: how to efficiently turn massively large data into valuable information and meaningful knowledge. Discrete mathematics produces a significant collection of powerful methods, including mathematical tools for understanding and managing very high-dimensional data, inference systems for drawing sound conclusions from large and noisy data sets, and algorithms for scaling computations up to very large sizes. Discrete mathematics is the mathematical language of data science, and as such, its importance has increased dramatically in recent decades.

Chapter 2

Set Theory

2.1 Definition of a set

Definition 2.1.1. A set is a well-defined collection of distinct objects .

Example 2.1.1. The set of all first year students at Karatina University

Example 2.1.2. The set of all letters in the English Alphabet.

Example 2.1.3. The set of all positive whole numbers.

2.2 Notations

Each of the objects in the set is called a member or an element of the set. The objects themselves can be almost anything. Books, cities, numbers, animals, flowers, etc. Elements of a set are usually denoted by lower-case letters. While sets are denoted by upper case letters of the English alphabet.

The symbol $\in$ indicates the membership in a set.

The symbol $\in$ is read “is a member of ” or “is an element of ”.

We write $a \in A$ to denote that a is an element of the set A .

The notation $a \notin A$ denotes that a is not an element of the set A .

2.2.1 Specifying Sets

1. **Descriptive Form:** Involves stating in words the elements of the set.

Examples 2.2.1. i) $A =$ Set of first five natural numbers.

ii) $B =$ Set of positive even integers less than or equal to fifty.

iii) $C =$ Set of positive odd integers.

2. **The Roster Method:** In this method, one lists all the elements in the set between a pair of braces.

Example 2.2.1. The set V of all vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$.

Example 2.2.2. The set O of odd positive integers less than 10 can be expressed by $O = \{1, 3, 5, 7, 9\}$.

Remark. In this method the order in which the elements are listed is immaterial, and is used for small sets.e.g. $O = \{1, 3, 5, 7, 9\} = \{3, 5, 1, 9, 7\}$.

Repetition of objects is not allowed in a set also.

Sometimes the roster method is used to describe a set without listing all its members. Some members of the set are listed, and then ellipses (...) are used when the general pattern of the elements is obvious.

Example 2.2.3. The set of positive integers less than 100 can be denoted by $\{1, 2, 3, \dots, 99\}$.

3. **Set Builder Notation:** In this method, we characterize all those elements in the set by stating the property or properties they must have to be members.

Example 2.2.4. The set O of all odd positive integers less than 10 can be written as $O = \{x \mid x \text{ is an odd positive integer less than } 10\}$

The vertical bar $\mid$ is read as “such that”.. The colon $:$, can also be used instead.

Alternatively, let $\mathbb{Z}^+$ denote the set of positive integers, then we can write

$O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}$ for the set O of all odd positive integers less than 10

We often use this type of notation to describe sets when it is impossible to list all the elements of the set. For instance, the set $\mathbb{Q}^+$ of all positive rational numbers can be written as:

$$\mathbb{Q}^+ = \left\{ x \in \mathbb{R} \mid x = \frac{p}{q}, \text{ for some positive integers } p \text{ and } q \right\}.$$

2.3 Some important sets used in Mathematics

- i) $\mathbb{N} = \{1, 2, 3, \dots\}$: The set of Natural Numbers. A natural number is an integer greater than 0. They are the positive whole numbers.
- ii) $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$: the set of integers. An integer (from the Latin integer meaning "whole") is a number that can be written without a fractional component. They are the positive whole numbers, negative whole numbers and zero. NB: Zero is neither positive nor negative, it is considered neutral.
- iii) $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$. This is the set of positive integers.
- iv) $\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, \text{ and } q \neq 0 \right\}$. The set of Rational numbers. A rational number is a number that can be expressed as the quotient or fraction p/q of two integers, a numerator p and a non-zero denominator q . Since q may be equal to 1, every integer is a rational number.
- v) $\mathbb{R}$, the set of real numbers. The real numbers include all the rational numbers, and all the irrational numbers.
- vi) $\mathbb{R}^+$, the set of positive real numbers.
- vii) $\mathbb{C}$, the set of complex numbers.

Recall the notation for intervals of real numbers. When a and b are real numbers with $a < b$, we write:

$[a, b] = \{x : a \leq x \leq b\}$, called a closed interval.

$[a, b) = \{x : a \leq x < b\}$, called half-closed or half-open interval.

$(a, b] = \{x : a < x \leq b\}$, called half-closed or half-open interval.

$(a, b) = \{x : a < x < b\}$, called an open interval.

2.4 Subsets

Definition 2.4.1. The set A is a subset of set B if and only if every element of A is also an element of B . We use the notation $A \subseteq B$ to indicate that A is a subset of the set B . We also say that A is contained in B .

Rules for determining whether one set is a subset of another:

To show that $A \subseteq B$, show that if x belongs to A then x also belongs to B .

To show that A is not a subset of B , denoted, $A \not\subseteq B$, find a single $x \in A$ such that $x \notin B$.

Remark. Every set is a subset to itself, i.e. $A \subseteq A$.

If A is contained in B , then we may also state that B contains A and write $B \supseteq A$.

Example 2.4.1. $\{2, 4, 6\} \subseteq \{2, 4, 6, 8, 10\}$

Example 2.4.2. $\{a\} \subseteq \{a, b, c\}$

Example 2.4.3. $\{a, d\} \not\subseteq \{a, b, c\}$

Example 2.4.4. $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{R} \subseteq \mathbb{C}$

2.4.1 Proper subset

Definition 2.4.2. A set A is called proper subset of the set B if A is subset of B and B is not a subset A i.e., A is said to be a proper subset of B if every element of A belongs to the set B , but there is at least one element of B , which is not in A . If A is a proper subset of B , then we denote it by $A \subset B$.

2.4.2 Equal sets

If A and B are sets such that every element of A is an element of B and every element of B is an element of A , then A and B are equal (Identical) sets, denoted $A = B$.

To show two sets A and B are equal, show that $A \subseteq B$ and $B \subseteq A$.

2.4.3 Super set

If A is subset of B , then B is called a superset of A .

2.4.4 The Null set or the Empty set

Definition 2.4.3. The set with no elements is called an empty set or null set denoted by the symbol $\emptyset$ or $\{\}$.

2.4.5 Singleton Set

Definition 2.4.4. A set having only one element is called a singleton.

Example 2.4.5. $A = \{0\}$

Example 2.4.6. $A = \{\emptyset\}$

2.5 Cardinality of a set

Definition 2.5.1. The number of elements in a set is the cardinality of that set. The cardinality of the set A is often notated as $|A|$ or $n(A)$.

Definition 2.5.2. Let A be a set. If there are exactly n distinct elements in A where n is a non-negative integer, we say that A is a **finite set** and that n is the cardinality of A .

Example 2.5.1. Let A be the set of odd positive integers less than 10. Then $|A| = 5$.

Example 2.5.2. Let S be the set of letters in the English alphabet. Then $|S| = 26$.

Example 2.5.3. Because the null set has no elements, it follows that $|\emptyset| = 0$.

Definition 2.5.3. A set is said to be infinite if it is not finite.

Example 2.5.4. The set of positive integers is infinite.

2.5.1 Number of subsets for a finite set

If a set contains n elements, then the number of subsets of the set is 2^n .

Example 2.5.5. List all the subsets of the set $A = \{1, 2, 3\}$.

Solution:

The subset of A containing no elements - $\emptyset$

The subset of A containing one element each - $\{1\}, \{2\}, \{3\}$

The subset of A containing two elements each - $\{1, 2\}, \{1, 3\}, \{2, 3\}$

The subset of A containing three elements - $\{1, 2, 3\}$

Therefore, all possible subsets of A are $\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$.

These are $8 = 2^3$ in total.

The table below gives a summary on the list of all possible number of subsets for a given finite set.

Table 2.1: **Listing Subsets**

Set	List of all subsets	Number of subsets
$\emptyset$	$\emptyset$	$2^0 = 1$
$\{a\}$	$\emptyset, \{a\}$	$2^1 = 2$
$\{a, b\}$	$\emptyset, \{a\}, \{b\}, \{a, b\}$	$2^2 = 4$
$\{a, b, c\}$	$\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}$	$2^3 = 8$
$\{a, b, c, d\}$		$2^4 = 16$
$\{a, b, c, d, e\}$		$2^5 = 32$
...	...	...
$\{a, b, c, \dots\}$		2^n

Exercise 1:

- List the proper subsets of the set $\{\alpha, \beta, \omega\}$
- List all the subsets of the set $\{\mathbb{N}, \mathbb{Z}, \mathbb{R}, \mathbb{C}\}$
- List all the subsets with two elements of the set $\{a, b, c, d, e\}$

2.5.2 The power set

Definition 2.5.4. Given a set S , the power set of S is the set of all subsets of the set S . The power set of S is denoted by $\mathcal{P}(S)$.

Example 2.5.6. What is the power set of the set $D = \{a, b, c\}$

Solution: The power set $\mathcal{P}(D) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$

Example 2.5.7. What is the power set of the empty set?

Exercise 2:

- i) What is the power set of the empty set?
- ii) What is the power set of the set $\{\emptyset\}$?

Remark. If a set has n elements, then its power set has 2^n elements.

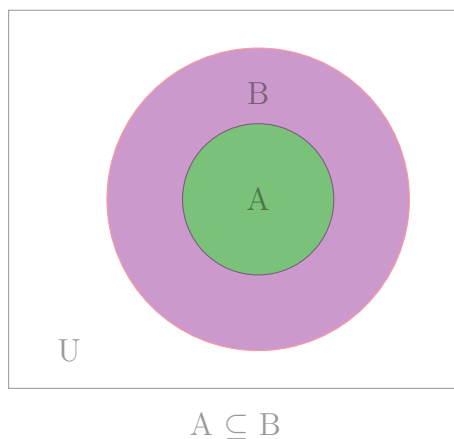
2.5.3 The Universal set

Definition 2.5.5. A Universal Set is the set of all elements under consideration, usually denoted by U . All other sets are subsets of the universal set.

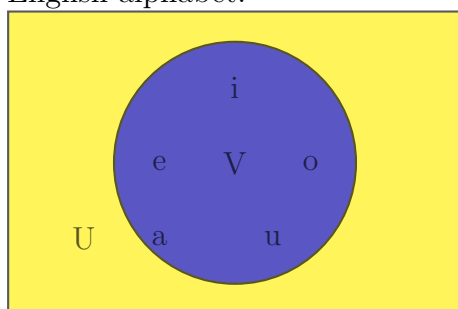
2.6 Venn Diagrams

A Venn Diagram is a pictorial representation of the relationships between sets. In Venn diagrams the universal set U , which contains all the objects under consideration, is represented by a rectangle. (Note that the universal set varies depending on which objects are of interest.) Inside this rectangle, circles or other geometrical figures are used to represent sets. Sometimes points are used to represent the particular elements of the set.

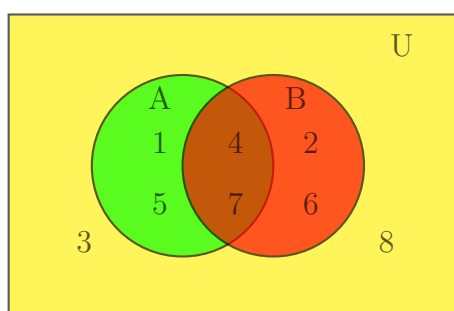
Example 2.6.1. Draw a Venn Diagram showing that set A is a subset of set B .



Example 2.6.2. Draw a Venn diagram that represents V , the set of vowels in the English alphabet.



Example 2.6.3. Draw a Venn diagram that represents the relationship between the sets: $U = \{1, 2, 3, \dots, 8\}$, $A = \{1, 4, 5, 7\}$ and $B = \{2, 4, 6, 7\}$.



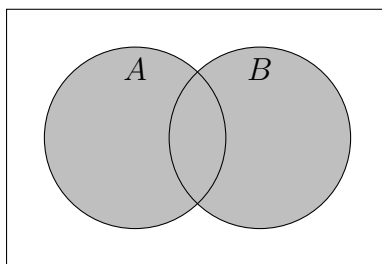
2.7 Set Operations

2.7.1 Union of sets

Let A and B be sets. The union of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in both.

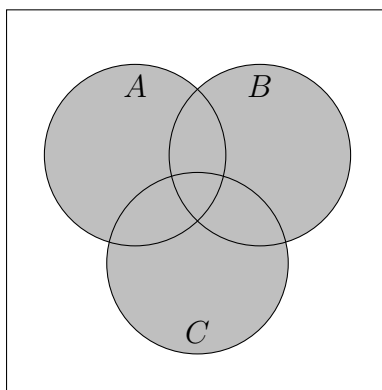
In symbols we write: $A \cup B = \{x : x \in A \text{ or } x \in B\}$

Venn Diagram for the union of two sets:



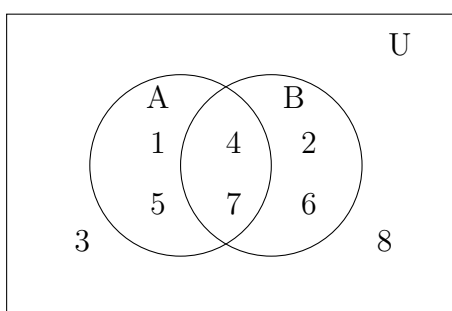
$A \cup B$ shaded

Venn Diagram for the union of three sets:



$A \cup B \cup C$ shaded

Example 2.7.1. Given $U = \{1, 2, 3, \dots, 8\}$, $A = \{1, 4, 5, 7\}$ and $B = \{2, 4, 6, 7\}$ find $A \cup B$.



$$A \cup B = \{1, 2, 4, 5, 6, 7\}$$

General case:

If $A_1, A_2, \dots, A_n$ denote sets, then the union of these sets denoted by $\bigcup_{i=1}^n A_i$ is defined as

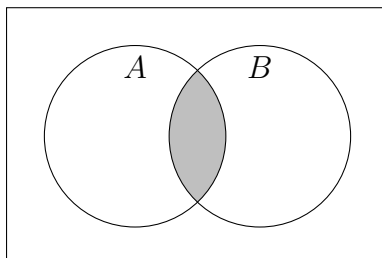
$$\bigcup_{i=1}^n A_i = \{x : x \in A_i \text{ for at least one set } A_i\}.$$

2.7.2 Intersection of sets

Let A and B be sets. The intersection of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .

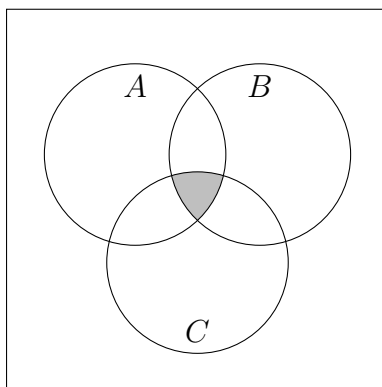
In symbols we write: $A \cap B = \{x : x \in A \text{ and } x \in B\}$

Venn Diagram for the intersection of two sets:



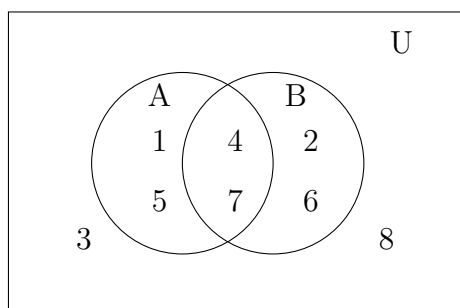
$A \cap B$ shaded

Venn Diagram for the intersection of three sets:



$A \cap B \cap C$ shaded

Example 2.7.2. Given $U = \{1, 2, 3, \dots, 8\}$, $A = \{1, 4, 5, 7\}$ and $B = \{2, 4, 6, 7\}$ find $A \cap B$.

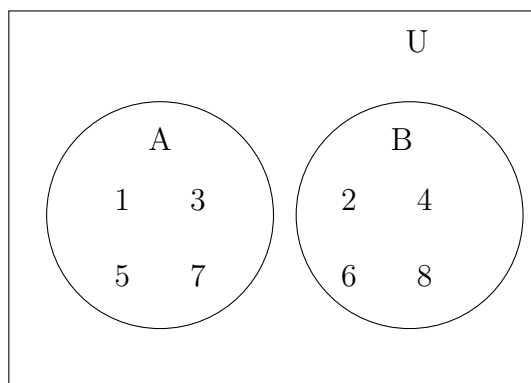


$$A \cap B = \{4, 7\}$$

Definition 2.7.1. Two sets are called disjoint if their intersection is the empty set.

Example 2.7.3. Let $A = \{1, 3, 5, 7\}$ and $B = \{2, 4, 6, 8\}$.

Then $A \cap B = \emptyset$ and A and B are disjoint sets

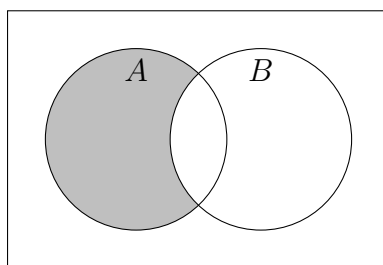


2.7.3 Set Difference

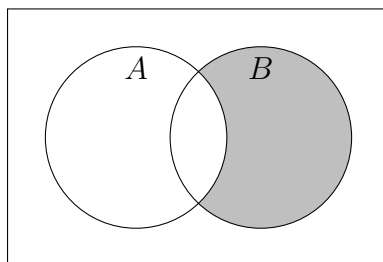
Let A and B be sets. The difference of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the complement of B with respect to A .

Remark. The difference of sets A and B is sometimes denoted by $A \setminus B$.

In symbols we write: $A - B = A \setminus B = \{x : x \in A \text{ and } x \notin B\}$



$A \setminus B$ Shaded

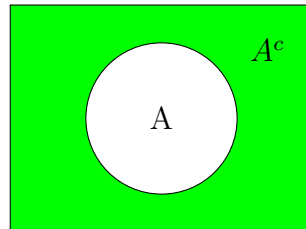


$B \setminus A$ Shaded

Definition 2.7.2. Let U be the universal set. The **complement** of the set A , denoted by $\bar{A}$ or A^c , is the complement of A with respect to U . Therefore, the complement of the set A is $U - A$.

An element belongs to A if and only if $x \notin A^c$. This tells us that:

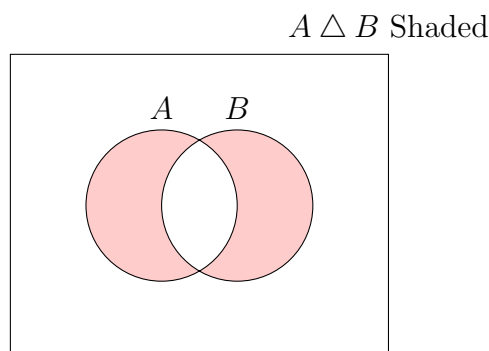
$$\bar{A} = A^c = \{x \in U : x \notin A\}$$



Venn Diagram for A^c

2.7.4 Symmetric Difference

The symmetric difference of A and B , denoted by $A \oplus B$ or $A \triangle B$, is the set containing those elements in either A or B , but not in both.



Remark. $A \oplus B = A \triangle B = (A \cup B) - A \cap B = (A - B) \cup (B - A)$

Example 2.7.4. Let the universal set be the set $U = \{a, b, c, d, e, f, g\}$ and let $A = \{a, c, e, g\}$ and $B = \{d, e, f, g\}$. Find $A \cup B$, $A \cap B$, $B - A$, A^c and $A \triangle B$.

Solution:

$$A \cup B = \{a, c, d, e, f, g\}$$

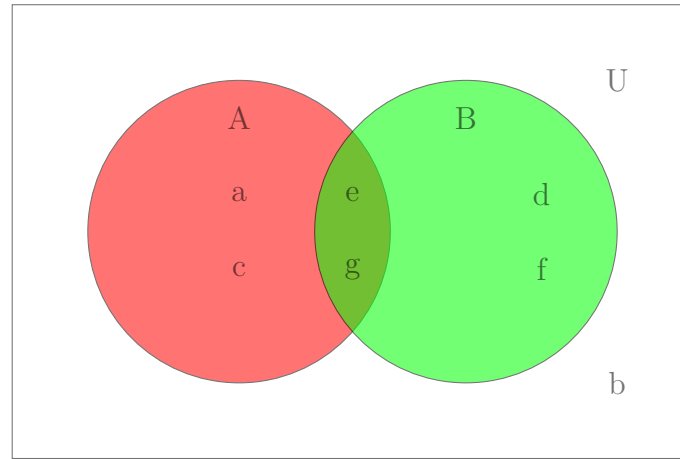
$$A \cap B = \{e, g\}$$

$$B - A = \{d, f\}$$

$$A^c = \{b, d, f\}$$

$$A \triangle B = \{a, c, d, f\}$$

On a Venn diagram the information can be illustrated as below:



2.7.5 Fundamental Products

Consider n distinct sets $A_1, A_2, \dots, A_n$.

A fundamental product of the sets is a set of the form $A_1^* \cap A_2^* \cap \dots \cap A_n^*$ where $A_i^* = A_i$ or $A_i^* = A_i^c$

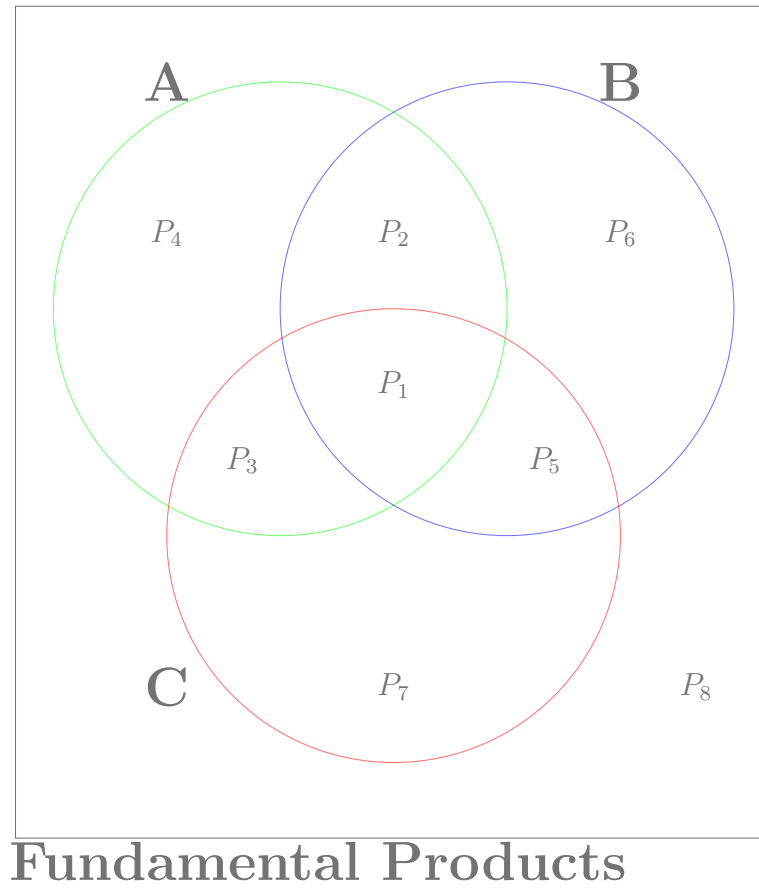
We note that:

- (i) There are $m = 2^n$ such fundamental products.
- (ii) Any two such fundamental products are disjoint.
- (iii) The universal set U is the union of all fundamental products.

Example 2.7.5. Consider three sets A, B, C . The following are the $8 = 2^3$ fundamental products of the sets:

$$\begin{aligned} P_1 &= A \cap B \cap C, & P_2 &= A \cap B \cap C^c, & P_3 &= A \cap B^c \cap C, & P_4 &= A \cap B^c \cap C^c, \\ P_5 &= A^c \cap B \cap C, & P_6 &= A^c \cap B \cap C^c, & P_7 &= A^c \cap B^c \cap C, & P_8 &= A^c \cap B^c \cap C^c \end{aligned}$$

The eight products correspond precisely to the eight disjoint regions in the Venn diagram of sets A, B, C as indicated by the labeling of the regions in the figure below:



2.7.6 Counting Elements in Finite Sets

The notation $n(S)$ or $|S|$ will denote the number of elements in a set S .

Cardinality of union of two sets: Suppose A and B are finite sets. Then

- i) $A \cup B$ is finite and if A and B are disjoint, then $n(A \cup B) = n(A) + n(B)$
- ii) $n(A - B) = n(A) - n(A \cap B)$
- iii) $A \cup B$ and $A \cap B$ are finite and $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. This is called the **Inclusion–Exclusion Principle**.

Cardinality of union of three sets: Suppose A , B and C are finite sets. Then $A \cup B \cup C$ is finite and

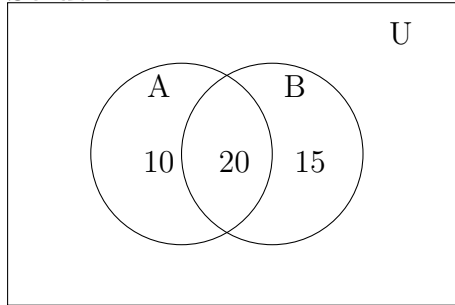
$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C).$$

Example 2.7.6. Suppose a list A contains the 30 students in a mathematics class, and a list B contains the 35 students in an English class, and suppose there are 20 names

on both lists. Find the number of students:

- (a) only on list A(Soln: 10 students)
 (b) only on list B(Soln: 15 students)
 (c) on list A or B (or both).....(Soln: 45 students)
 (d) on exactly one list.....(Soln: 25 students)

Solution:



Example 2.7.7. Out of 30 students in a dormitory, 15 take an art course, 8 take a biology course and 6 take a chemistry course. It is known that 3 students take all the three courses. Show that 7 or more students take none of the courses..

2.8 Set identities

An identity is an equation that is universally true for all elements in some set. For example, the equation $a + b = b + a$ is an identity for real numbers because it is true for all real numbers a and b . The collection of set properties in the next theorem consists entirely of set identities. That is, they are equations that are true for all sets in some universal set.

1. Commutative laws

a) $A \cup B = B \cup A$

b) $A \cap B = B \cap A$

2. Associative laws

a) $A \cup (B \cup C) = (A \cup B) \cup C$

b) $A \cap (B \cap C) = (A \cap B) \cap C$

3. Distributive laws

a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

4. Identity laws

a) $A \cap U = A$

b) $A \cup \emptyset = A$

5. Complement laws

a) $A \cup A^c = U$

b) $A \cap A^c = \emptyset$

6. Complementation law or the Double Complement Law: $(A^c)^c = A$

7. Idempotent laws :

a) $A \cup A = A$

b) $A \cap A = A$

8. Domination laws or the Universal Bound Laws:

a) $A \cup U = U$

b) $A \cap \emptyset = \emptyset$

9. De Morgan's laws

a) $(A \cap B)^c = A^c \cup B^c$

b) $(A \cup B)^c = A^c \cap B^c$

10. Absorption laws

a) $A \cup (A \cap B) = A$

$$\text{b) } A \cap (A \cup B) = A$$

11. Complements of U and $\emptyset$

$$\text{a) } U^c = \emptyset$$

$$\text{b) } \emptyset^c = U$$

12. Set Difference Law: $A - B = A \cap B^c$

2.8.1 Principle of Duality

Suppose E is an equation of set algebra. The dual E^* of E is the equation obtained by replacing each occurrence of $\cup, \cap, U$ and $\emptyset$ in E by $\cap, \cup, \emptyset$ and U , respectively. The pairs in the set identities are all duals of each other. It is a fact of set algebra, called the principle of duality, that if any equation E is an identity then its dual E^* is also an identity.

Example 2.8.1. The dual of $(U \cap A) \cup (B \cap A) = A$ is $(\emptyset \cup A) \cap (B \cup A) = A$

2.8.2 Proving Set Identities

Let X and Y be sets. To prove that $X = Y$:

$$\text{i) } X \subseteq Y$$

$$\text{ii) } Y \subseteq X$$

Example 2.8.2. Prove that for all sets A and B , $(A \cup B)^c = A^c \cap B^c$.

Solution: We need to show that : $(A \cup B)^c \subseteq A^c \cap B^c$ and $A^c \cap B^c \subseteq (A \cup B)^c$.

To show that $(A \cup B)^c \subseteq A^c \cap B^c$, it means we show that for all x , if $x \in (A \cup B)^c$ then $x \in A^c \cap B^c$.

Suppose $x \in (A \cup B)^c$. Then by definition of complements $x \notin A \cup B$.

But to say that $x \notin A \cup B$ means that : it is false that (x is in A or x is in B).

This is equivalent to saying that: x is not in A and x is not in B , which can be written as $x \notin A$ and $x \notin B$.

It follows, by definition of intersection, that $x \in A^c \cap B^c$.

Thus $(A \cup B)^c \subseteq A^c \cap B^c$.

Next we show that $A^c \cap B^c \subseteq (A \cup B)^c$.

We must show that; for all $x \in A^c \cap B^c$ then $x \in (A \cup B)^c$.

Suppose $x \in A^c \cap B^c$.

By definition of intersection, $x \in A^c$ and $x \in B^c$, and by definition of complement, $x \notin A$ and $x \notin B$. In other words, x is not in A and x is not in B . This implies that "it is false that (x is in A or x is in B), which can be written as $x \notin A \cup B$ by definition of unions.

Hence, by definition of complement $x \in (A \cup B)^c$.

It thus follows that $A^c \cap B^c \subseteq (A \cup B)^c$.

Conclusion: Since both set containments have been proved, $(A \cup B)^c = A^c \cap B^c$ by definition of set equality.

Example 2.8.3. Use set builder notation and logical equivalences to establish the De Morgan law $(A \cap B)^c = A^c \cup B^c$

$$\begin{aligned}
 (A \cap B)^c &= \{x : x \notin A \cap B\} && \text{by definition of complement} \\
 &= \{x : \text{it is false that } (x \in A \cap B)\} && \text{by definition of does not belong symbol} \\
 &= \{x : \text{its false that } (x \in A \text{ and } x \in B)\} && \text{by definition of intersection} \\
 &= \{x : \text{its false that } (x \in A) \text{ or its false that } (x \in B)\} && \text{by logical equivalences of statements} \\
 &= \{x : x \notin A \text{ or } x \notin B\} && \text{by definition of does not belong symbol} \\
 &= \{x : x \in A^c \text{ or } x \in B^c\} && \text{by definition of complements} \\
 &= \{x : x \in A^c \cup B^c\} && \text{by definition of union} \\
 &= A^c \cup B^c && \text{by meaning of set builder notation}
 \end{aligned}$$

Chapter 3

Relations

3.1 Introduction

Relationships between elements of sets are represented using the structure called a relation, which is just a subset of the Cartesian product of the sets. Relations can be used to solve problems such as determining which pairs of cities are linked by airline flights in a network, finding a viable order for the different phases of a complicated project, or producing a useful way to store information in computer databases.

3.2 Cartesian Product of sets

Definition 3.2.1. An ordered pair of elements a and b , where a is designated as the first element and b as the second element, is denoted by (a, b) . In particular, $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Definition 3.2.2. Consider two arbitrary sets A and B . The set of all ordered pairs (a, b) where $a \in A$ and $b \in B$ is called the product, or Cartesian product, of A and B usually denoted by $A \times B$, which is read “A cross B.”

By Definition, we write, $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$